Dams mark drought vulnerability rather than buffer it, and storage cannot offset a falling water supply in Chile

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Abstract

Expanding reservoir storage is the reflexive response to intensifying drought. Yet whether dams build resilience or merely mark where vulnerability already concentrates has remained untested under sustained drying, for want of a credible counterfactual. Here we treat reservoir presence as a basin-level intervention across central Chile during its ongoing megadrought, using statistical matching to compare each dammed basin against undammed basins that share its climate, size, and terrain, and asking whether the two respond differently to the same drought. Across every operational pathway, meteorological-to-streamflow drought transmission, evapotranspiration, cropland-area change, and the direct allocation of consumptive water rights, the reservoir effect brackets zero once siting is matched out. The cleanest test compares regulated reaches below each dam against physically unregulated reaches above it: the apparent buffering is just as strong upstream as downstream and disappears when basins are compared within the same climate, the signature of an aridity gradient rather than of regulation. Positive controls confirm the design would have detected a real buffering effect had one existed. These tests concern the annual-to-multiyear timescale of structural drought, not the sub-seasonal buffering that reservoirs are also built to provide. Meanwhile the whole storage band drifts downward while its seasonal refill amplitude shows no detectable change, pointing to a falling water supply rather than a degraded buffer. Reservoirs in Chile thus mark pre-existing, location-determined vulnerability rather than create resilience against multi-year drought, even if they still perform the intra-annual seasonal buffering they are primarily built for. Under structural aridification, expanding storage is a supply-blind response to a supply problem; adaptation must shift toward managing demand, reallocating water, and recharging aquifers.

Keywords: reservoir effect, drought propagation, socio-hydrology, causal inference, megadrought, Chile

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1. Introduction

As the world dries, governments instinctively build more dams. Reservoirs are the dominant structural response to drought, and in the short term they work: more storage raises water availability and lowers the frequency, severity and duration of shortage (Di Baldassarre et al., 2018). Yet a socio-hydrological view warns that the same infrastructure can be self-defeating over a dam's planning horizon. Greater supply invites agricultural, urban and industrial expansion, so demand grows to meet and then exceed each new increment of storage (the supply-demand cycle); and prolonged reliance on abundant stored water erodes the incentive to adapt, deepening dependence and magnifying the damage when a drought finally outlasts the reservoir (the reservoir effect) (Di Baldassarre et al., 2018). Consistent with this, global water demand has outpaced storage for decades, the storage returns of new dams are diminishing worldwide (Li et al., 2023), and most reservoirs now fail to refill under intensifying drought (Wang et al., 2024). Whether these feedbacks dominate, or are outweighed by the protection dams provide, remains debated (Kellner, 2021).

The obstacle is that the reservoir effect for drought has been described far more than it has been measured. Reservoirs are not sited at random: they are built where irrigation demand and aridity already concentrate, so dammed basins differ systematically from undammed ones before any drought strikes. The short-term hydrological benefit is documented, as regulation shortens downstream hydrological drought and lengthens the lag from meteorological drought (Wu et al., 2017, 2018), though it can also redistribute scarcity in space and time (López-Moreno et al., 2009). But the long-term, demand side is far less constrained, and separating a genuine reservoir effect from the siting that accompanies it requires a defensible counterfactual, namely what a drought's impact *would have been* in a basin had the reservoir not been built (Di Baldassarre et al., 2018). Prior dammed-versus-undammed and before-after comparisons cannot supply it.

Chile is an acute testbed. Since 2010 its central regions have endured a *megadrought*, an uninterrupted run of dry years now shading into structural aridification as precipitation falls and atmospheric evaporative demand rises (Garreaud et al., 2025, 2017; Zambrano et al., 2025). Because the 1981 Water Code makes water-use rights perpetual private property that the state cannot readily curtail during scarcity, the policy reflex has been supply-side: a national programme of new dams and inter-basin transfers, alongside market incentives that concentrated water in high-value, water-demanding export crops (Barría et al., 2021; Reade Malagueño and D'Odorico, 2024). Chile thus exhibits, in compressed form, every ingredient of the reservoir effect: an extreme and persistent forcing, a supply-side response, and demand expansion under weak adaptive feedback.

Here we ask whether reservoirs build resilience or merely mark, and postpone, vulnerability under prolonged drought. We treat the presence of a major reservoir as a basin-level intervention and construct the missing counterfactual in two steps. First, we compare each dammed basin only against undammed basins that share its climate and landscape, and condition on the drought actually experienced, asking whether dammed and comparable undammed basins convert the same meteorological deficit into different impacts. Second, within each dammed basin we compare regulated reaches below the dam against physically unregulated reaches above it, which share climate, geology and siting but not the reservoir. Together these turn the socio-hydrological reservoir-effect hypothesis into a falsifiable, causally framed test, in the setting where it matters most.

2. Methods

2.1. Study area and design

We study continental Chile (18–44° S), a strong north–south aridity gradient spanning hyper-arid, Mediterranean, and humid-temperate climates, over 2005–2024, the window in which reservoir storage, meteorological forcing, and satellite outcomes jointly exist. The design is a matched comparison of **dammed versus undammed sub-watersheds**, with reservoir presence treated as a basin-level intervention. Because the reservoirs in most treated basins (18 of the 24) were commissioned before the storage record begins, treatment is effectively time-invariant over the window and a staggered-adoption (commissioning) event study is infeasible; identification therefore rests on a matched cross-section combined with conditioning on meteorological forcing rather than on calendar time.

The central estimand is the operational effect of a reservoir on drought outcomes *conditional on the covariates that govern where reservoirs are sited*, that is, whether dammed basins respond to the same drought forcing differently from hydroclimatically comparable undammed basins. This isolates reservoir *operation* from reservoir *siting* (Di Baldassarre et al., 2018).

2.2. Spatial units and treatment assignment

The analysis grain is the DGA BNA **sub-watershed (subcuenca)** (467 units), chosen over the coarser cuenca grain (101 units) for power and common support: reservoirs sit in Chile's largest main-stem basins, and the finer grain supplies far more controls inside the treated size range (detailed in the archived analysis repository). A subcuenca is **treated** if it contains 1 of the 26 monitored reservoirs by point-in-polygon assignment, giving **24 treated units**. An untreated subcuenca that lies within a cuenca containing a reservoir is flagged as **contaminated** (it shares the regulated main stem and is not a clean counterfactual) and removed (45 units), leaving **398 clean controls**. This contamination rule defuses the most direct downstream-spillover (SUTVA) violation; residual adjacency and shared-aquifer spillovers between neighbouring subcuencas are carried as an identification caveat and checked at cuenca grain.

2.3. Data

Reservoir storage and treatment. Monthly storage for 26 monitored reservoirs (the DGA record runs 2005-01 to 2026-04; we analyse the complete calendar years 2005–2024) with a per-reservoir capacity column (`max_level_hm3`); storage and capacity share units (hm^3), so percent-of-capacity (`100 × level / max_level_hm3`) is the cross-reservoir-comparable state variable. Static attributes (commissioning year, use, size class) come from the national dam registry.

Meteorological forcing. The Standardized Precipitation-Evapotranspiration Index (SPEI) at a 12-month accumulation, derived from CHIRPS precipitation and CHIRTS temperature and standardized on the 1991–2020 climatological baseline (Vicente-Serrano et al., 2010), is the primary exogenous dose; CHIRPS-derived standardized indices are validated for Chilean drought monitoring, including ENSO-modulated variability across the country (Meza, 2013; Oertel et al., 2020; Zambrano et al., 2017). We use the 12-month accumulation because it matches the annual, snowmelt-fed hydrological cycle of central Chile (winter precipitation stored as snow and released through the growing season) and so captures the inter-seasonal carryover that irrigation reservoirs are built to bridge, at the scale at which multi-year vulnerability (rather than sub-seasonal operation) is expressed. The treatment estimates are insensitive to this choice: re-estimating the forcing-conditioned effect with 6- and 24-month accumulations leaves it materially unchanged (archived analysis code). By construction the 12-month window does not resolve short-term (1–3 month) operational buffering, which reservoirs are documented to provide (Wu et al., 2017, 2018) and which we flag as a boundary of our conclusions (Discussion). Conditioning on forcing (estimating effects on the deficit-to-impact *slope* rather than on levels) is how we break the collinearity between the 2010-onset megadrought and the treatment clock.

Hydrological drought. Daily streamflow from the CR2 network (2000–2020) is aggregated to the Standardized Streamflow Index at 12 months [SSI-12; Vicente-Serrano et al. (2012); Lema et al. (2025)], the regulated-flow analogue of SPEI-12, used as the outcome in the within-basin placebo (below). Gauges are classified as upstream or downstream of each dam from an SRTM digital elevation model.

Ecological and land-use outcomes. (i) annual cropland area from MapBiomas Chile Collection 2 (30 m, 1999–2024); (ii) actual evapotranspiration from MODIS MOD16 (8-day, ~500 m, 2001–2024), for whole basins and for an orchard stratum delineated from the CIREN/ODEPA cadastre.

Water rights. The DGA consolidated registry of water-use rights (*Derechos de Aprovechamiento de Aguas*), giving for each right its grant date, consumptive/non-consumptive type, surface/groundwater nature, use, allocated flow (harmonized to l/s), and UTM capture point. We assign consumptive rights to subcuencas by point-in-polygon and build a per-basin annual panel of the cumulative allocated-rights stock, our direct demand-side outcome. Because the registry contains severe volume errors (a single physically impossible entry can dominate a basin's total, e.g. a caudal exceeding 200,000 m^3/s), the outlier-robust primary measure is rights *density* (count per 100 km^2); a volume series in which each individual consumptive flow is winsorized at the 99.5th percentile of the consumptive-flow distribution gives the same result, and the volume null is insensitive to the winsorization threshold across the 95th, 99th, 99.5th and 99.9th percentiles (permutation p from 0.39 to 0.21; Supplementary Table S8). Both are reported (Supplementary Fig. S6).

Matching covariates. Baseline aridity (long-term mean annual P/PET over the 1991–2020 WMO normal), watershed area, mean elevation (SRTM), and Köppen–Geiger climate class, extracted per subcuenca by zonal statistics.

2.4. Matched control construction

We estimate the average treatment effect on the treated (ATT) using **entropy balancing** (`WeightIt`, `method = "ebal"`) exact-matched within Köppen main group, balancing **log(area)** and **mean elevation** (see the archived analysis repository). Balancing on size and elevation rather than latitude is deliberate: latitude and the climate class fight each other across the Andes–Patagonia span, and substituting SRTM elevation raised the effective control sample size from 9 to 91 while balancing latitude for free.

The procedure retains **21 of 24 treated subcuencas** (three high-Andes units fall outside local elevation common support) and achieves essentially perfect balance (standardized mean differences for log(area) and elevation fall from 1.18 / 0.37 to 0.000, variance ratios 1), with an **effective control sample size 91** out of 244 in-window controls. Baseline aridity is a genuine confound (dammed basins are markedly more arid: median P/PET 0.26 vs 0.80 unweighted) but is *not* used as a balance target, because hard-balancing it collapses the ESS from 91 to 19; instead the weighting reduces the aridity imbalance to a residual SMD 0.17 (still above the 0.1 target), which the doubly-robust outcome model absorbs. Because this residual is real, the outcome-model adjustment for aridity is *load-bearing* rather than cosmetic and extrapolates across a genuine regime gap: the reported cross-sectional ATTs use a **linear** log-aridity adjustment, adding a quadratic aridity term leaves the qualitative conclusions unchanged (archived robustness), and restricting to the aridity-overlap subset (all 21 treated basins and 158 of 244 controls fall inside it) leaves the streamflow transmission-slope gap essentially unchanged (-0.17 vs -0.18) while shrinking the water-rights ATT to a value that now straddles zero, so the parametric adjustment is not masking a true effect (Supplementary Table S6). To rule out that the nulls are artefacts of this parametric functional form, we additionally re-fit entropy balancing *within* the aridity-overlap subset and estimate every cross-sectional outcome with the weighting-only estimator, which carries no outcome-model aridity term and so performs no extrapolation across the regime gap. On this fully non-parametric matching specification, cropland-area expansion, orchard-ET buffering and the water-rights count all remain null (Supplementary Table S7); the winsorized water-rights volume, which behaves differently here, is treated under spatial inference below. Crucially, the study's decisive evidence, the within-basin upstream/downstream placebo, the within-region transmission-slope comparison, and randomization inference, does **not** use the outcome-model aridity adjustment at all, so the central nulls do not hinge on extrapolating across the aridity gap. Robustness to the matching method is assessed with coarsened exact matching (CEM) and 1:2 nearest-neighbour (Mahalanobis) matching on the identical common-support sample; entropy balancing and NN agree on all 21 treated, while CEM additionally prunes the 4 most extreme large/high basins, so all treatment effects are reported with and without those units.

2.5. Causal estimands and estimators

Doubly-robust matched ATT. Each outcome is estimated three ways on the same sample, comprising weighting-only (ebal-weighted difference in means), regression-only (covariate-adjusted g-computation), and the headline **doubly-robust** estimator (ebal weights *and* an outcome model in log(area), elevation, and log(aridity)), so functional-form sensitivity is visible. Continuous covariates are centred at the treated mean so the treatment coefficient reads as the ATT; standard errors are HC3-robust (ebal weights treated as fixed, the standard mildly anti-conservative simplification).

Forcing-conditioned transmission slope. To remove megadrought *exposure* (concentrated in exactly the arid basins where reservoirs sit), the outcome for the DR machinery is the per-basin **transmission slope**, the regression of an annual outcome (e.g. log evapotranspiration) on annual SPEI-12 over 2005–2024, a standardized, cross-basin drought-transmission elasticity. The vulnerability hypothesis predicts a *steeper* slope (more impact per unit deficit) in dammed basins.

The two cross-sectional ATTs in Table 1 apply this machinery to (i) *cropland-area expansion*, the change in MapBiomas cropland area per basin over the window, and (ii) *reservoir ET buffering*, the per-basin transmission slope of log evapotranspiration on SPEI computed over **irrigated orchard cells** (MOD16 ET on the CIREN/ODEPA cadastre footprint), which measures directly whether consumptive water use over irrigated land responds more or less to deepening drought in dammed basins. A negative slope gap indicates buffering, a positive one induced-demand vulnerability; the outcome is null, so consumptive use per unit of irrigated land shows no differential change with the reservoir.

Forcing-interacted difference-in-differences. On the observed area and ET panels we fit

$$y_{it} = \beta\,(\mathrm{dammed}_i \times \mathrm{SPEI}_{it}) + \gamma\,\mathrm{SPEI}_{it} + \alpha_i + \tau_t + \varepsilon_{it},$$

with subcuenca (α_i) and year (τ_t) fixed effects and entropy-balancing weights. Because treatment is time-invariant, the dose is meteorological SPEI rather than calendar year: the year fixed effects absorb the common megadrought shock, and β is the differential deficit-to-impact slope between dammed and matched-control basins, robust to the siting-in-arid-central-Chile confound, which biases levels far more than slopes. The vulnerability hypothesis predicts $\beta > 0$.

Equivalence testing and minimum detectable effect. Because the headline results are nulls, we move beyond failure-to-reject and bound the operational effect. For each null we report a two one-sided test (TOST): the effect is declared equivalent to zero if its 90% confidence interval lies within a negligible region of ±25% of the baseline (untreated) transmission slope. We are not aware of an established hydrological or operational standard defining a "negligible" reservoir effect on drought transmission, so we anchor the margin in the granularity of operational drought classification rather than choosing it arbitrarily. Standardized drought indices are managed in discrete severity categories each 0.5 standardized units wide (McKee et al., 1993; Vicente-Serrano et al., 2010). Because the estimand is a slope, the absolute change it induces in the propagated index scales with the SPEI deficit, so the relevant test is at the most extreme drought actually observed. The largest SPEI-12 deficit in the panel is ≈ -2.0 (minimum -1.98), and 25% of the baseline transmission slope is $0.25 \times 0.585 \approx 0.146$ SSI per unit SPEI; the maximum absolute perturbation of the streamflow-drought index a 25% slope alteration can produce is therefore $0.146 \times 1.98 \approx 0.29$ SSI units, below the 0.5-unit width of a single severity category. A 25% slope change thus cannot move a basin across an operational drought-severity threshold even under the worst observed deficit, and is marginal for drought management. We stress that 25% of the transmission slope can nonetheless correspond to a non-trivial absolute water volume in a large, high-demand basin; the margin is anchored to the operational drought-severity classification, not to volumetric negligibility, which is why we also report the threshold-free MDE and rest the causal weight of the null on the within-basin placebo rather than on the equivalence verdict. We adopt $\pm 25\%$ as this deliberately strict, pre-specified margin and, because the equivalence verdict is nonetheless sensitive to the bound, also report the threshold-free MDE so that readers may apply any margin they consider material. The qualitative conclusion (only large effects are excluded; no outcome is equivalent at $\pm 25\%$) does not depend on the exact margin. The minimum detectable effect (at 80% power, two-sided $\alpha = 0.05$) and the equivalence interval are built from the **standard deviation of the permutation-null distribution**, not the analytic standard error, since cluster-robust SEs over-reject at ~21 clusters (archived analysis code). To confirm the null is informative rather than underpowered, we additionally run a **positive control**: a known buffering slope is injected into the treated units and the full estimator and randomization inference are re-run, verifying that the design recovers an effect of policy-relevant size and correctly flags none when none is injected.

Siting-confound decomposition. To locate where any apparent buffering comes from, we fit a ladder on the same panel: (1) a naive pooled slope gap (no weights, no fixed effects, what a conventional dammed-vs-undammed study reports); (2) unit + time fixed effects, unweighted; (3) the design estimator (ebal weights + fixed effects); (4) the design estimator with the baseline SPEI-to-outcome slope allowed to vary by Köppen region. It is important that these are distinct controls. The entropy balancing exact-matches *basin size and elevation* within Köppen main group; it does **not** balance the residual aridity gradient (which varies within B and C) nor impose a region-specific dose-response. The apparent buffering can therefore survive rungs (1) to (3), whose weighting leaves the within-region aridity gradient intact, and die only at rung (4), which lets each climate region have its own baseline transmission and so removes the between-region contrast that the matching does not. The collapse from (3) to (4) is the between-region aridity confound; the residual at (4) is the regulation estimate.

Within-basin upstream/downstream placebo. The primary selection-versus-effect discriminator contrasts the SPEI-12→SSI-12 transmission slope at regulated **downstream** reaches against physically unregulated **upstream** reaches of the *same* dammed basins, which share climate, geology, and selection history. A genuine regulation effect implies a downstream-specific dose-response that the upstream placebo does not reproduce; an attenuation equally present upstream is siting, not regulation. Of the 21 treated matched basins, 15 have both an upstream and a downstream gauge, giving near-national coverage rather than a single case study. The contrast is also split by Köppen region to test whether a regulation effect emerges in the high-supply south. Because upstream gauges sit at higher elevations than downstream gauges, we test the placebo's comparability assumption directly: we quantify the elevation gap and regress the per-gauge transmission slope on elevation among undammed control gauges, so that any elevation-driven regime difference is measured rather than assumed.

2.6. Inference

With only ~21 treated clusters, cluster-robust standard errors over-reject, so the primary inference is **randomization inference**: the treatment label is permuted among units within Köppen × aridity- tercile strata (999 permutations) and the observed statistic is compared against the resulting null distribution; cluster-robust confidence intervals are reported alongside for completeness. Because hydrological and land-use outcomes can be spatially autocorrelated, we assess residual spatial dependence for each outcome with Moran's I (great-circle distances, five-nearest-neighbour row-standardized weights, tested by 999 residual permutations). Where it is present, notably the water-rights outcomes ($I \approx 0.4$ to 0.6, $p = 0.001$; Supplementary Table S9), the unit-level stratified permutation can be anti-conservative because it does not preserve spatial structure. We therefore re-run the affected outcomes with a spatially-restricted permutation that swaps treatment at the cuenca-block level, so co-located subcuencas move together, and with cuenca-block-clustered confidence intervals; this spatially-valid inference governs where the naive and spatial tests disagree (Supplementary Table S9). Parallel pre-trends are checked with a dynamic event study (dammed × year, reference 2009), and residual confounding is probed with an aridity placebo that relabels the most-arid control tercile as pseudo-treated.

2.7. Software and reproducibility

All analyses are implemented in R within a `targets` pipeline (matched-set construction, estimators, and figures are individual targets), using `WeightIt/cobalt` for balancing and diagnostics, `fixest` for fixed-effects regression, `sf/terra/exactextractr` for the spatial extractions, and `renv` for package pinning. Every result in the manuscript corresponds to a named pipeline target; no figure or table is produced by hand.

3. Results

3.1. Reservoirs sit where water is already scarce

Reservoirs in Chile are built where drought bites hardest: our 21 dammed basins cluster in the arid centre and north, where irrigation demand and aridity already coincide, so a raw comparison of dammed against undammed basins would confuse the reservoir with its location. We separate the two with the design in Figure 1. First, we compare each dammed basin only with undammed basins that share its climate, size and elevation, statistical matching that keeps 21 of 24 dammed basins against an effective 91 of 244 controls (Supplementary Figs. S4 and S5), and we condition on the drought each basin actually experienced, so we compare how much impact a basin registers per unit of meteorological deficit rather than its raw level. Second, within each dammed basin we compare regulated reaches below the dam against physically unregulated reaches above it. A genuine reservoir effect must show up in these comparisons; across every pathway, it does not.

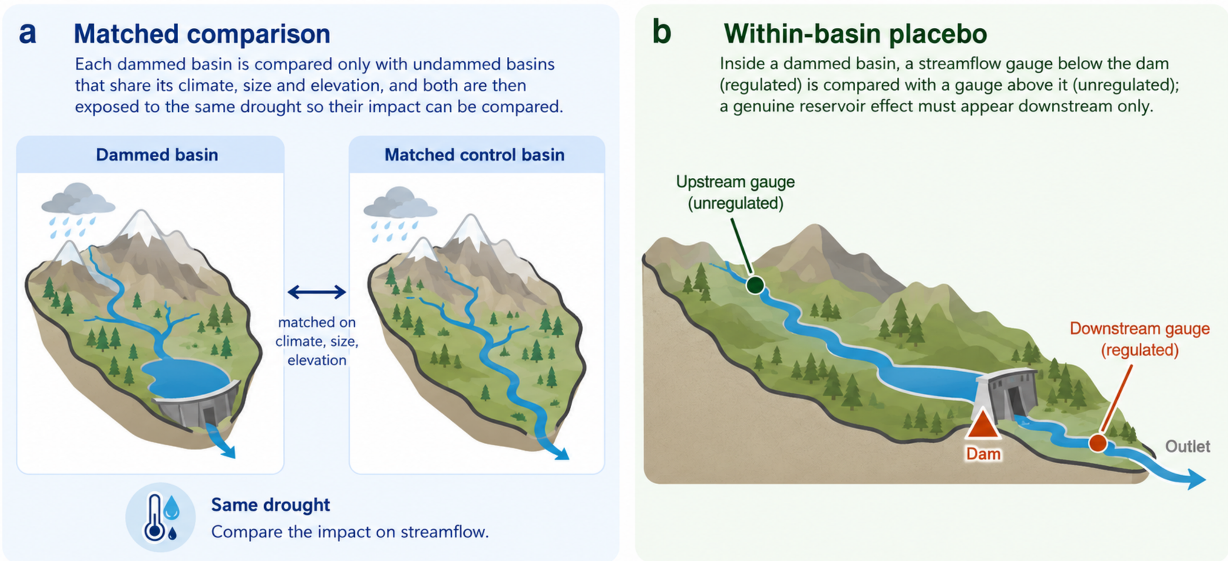

Figure 1: The two comparisons that separate the reservoir from its location. (a) Matched comparison: each dammed basin is compared only with undammed basins that share its climate, size and elevation, and both are then exposed to the same drought so their impact can be compared. (b) Within-basin placebo: inside a dammed basin, a streamflow gauge below the dam (regulated) is compared with a gauge above it (unregulated); a genuine reservoir effect must appear downstream only. Schematic; the full study-area map and matching diagnostics are in Supplementary Figs. S4 and S5.

3.2. The apparent buffering is the location, not the dam

The most direct and intuitive test is on streamflow. Meteorological drought (SPEI-12) propagates into streamflow drought (SSI-12) more gently in dammed basins, the visual signature of buffering (Figure 2, panel a). But this flattening tracks where dams sit, not what they do. If the dam were doing the buffering, transmission would flatten only below it; instead it flattens just as much at the unregulated *upstream* reaches of the same basins (differential slope -0.20 upstream versus -0.16 downstream; Figure 2, panel b). Because the upstream line flattens too, the cause must be the aridity of the location, not the reservoir. The national slope gap of -0.18 accordingly collapses to about -0.03 once dammed and control basins are compared within the same climate region, and in the high-supply south, where buffering should be easiest to see, the upstream and downstream gaps are statistically identical (-0.036 versus -0.034). None survives randomization inference ($p_{\text{perm}} = 0.39$). The apparent buffering is therefore the between-region aridity gradient, the textbook signature of a siting confound, not an effect of regulation; a step-by-step decomposition confirms the signal survives matching and fixed effects, which balance basin size and elevation but leave the within-region aridity gradient intact, and dies only when each climate region is allowed its own baseline transmission (Supplementary Fig. S3 and Table S4). Upstream and downstream reaches are not perfectly interchangeable, since upstream gauges sit higher and elevation predicts transmission among controls, but a bounding argument shows elevation cannot manufacture a downstream-specific buffering effect large enough to matter. The 0.54 km upstream-downstream elevation gap, scaled by the control elevation-transmission sensitivity (-0.21 per km), implies elevation alone would make the upstream slope about 0.11 more buffered than downstream, the same sign as, and larger in magnitude than, the observed -0.04 upstream-minus-downstream difference. Elevation therefore already over-explains the entire observed gap; attributing all of it to elevation leaves a residual downstream-specific buffering of at most about 0.07 (roughly 13% of the 0.53 baseline transmission slope), well inside the negligibility margin and far below the 46 to 58% of baseline the design can detect (Supplementary Table S2).

3.3. No reservoir effect survives across any operational pathway

The same answer holds across the other pathways by which reservoirs are thought to modify drought (Figure 3, Table 1). We measure each effect as the differential response of dammed and matched-control basins to the same meteorological deficit, and place them on a common axis: each estimate is divided by its own standard error, so effects on different natural scales (hectares, evapotranspiration, streamflow) can be read on one plot, and a value inside ± 2 is indistinguishable from no effect. Every interpretable estimate brackets zero: cross-sectional cropland-area expansion (-0.69 ha km^{-2}, 95% CI $[-2.4, 1.1]$) and evapotranspiration buffering (-0.014 $[-0.065, 0.037]$), and the forcing-conditioned slope gaps for cropland area ($p_{\text{perm}} = 0.91$) and orchard evapotranspiration ($p_{\text{perm}} = 0.24$). One outcome, whole-basin evapotranspiration, appears positive, but its pre-trends fail and it is confounded by residual aridity over barren land, so we flag it as uninterpretable rather than fold it silently into the nulls; at face value it points toward *vulnerability*, not

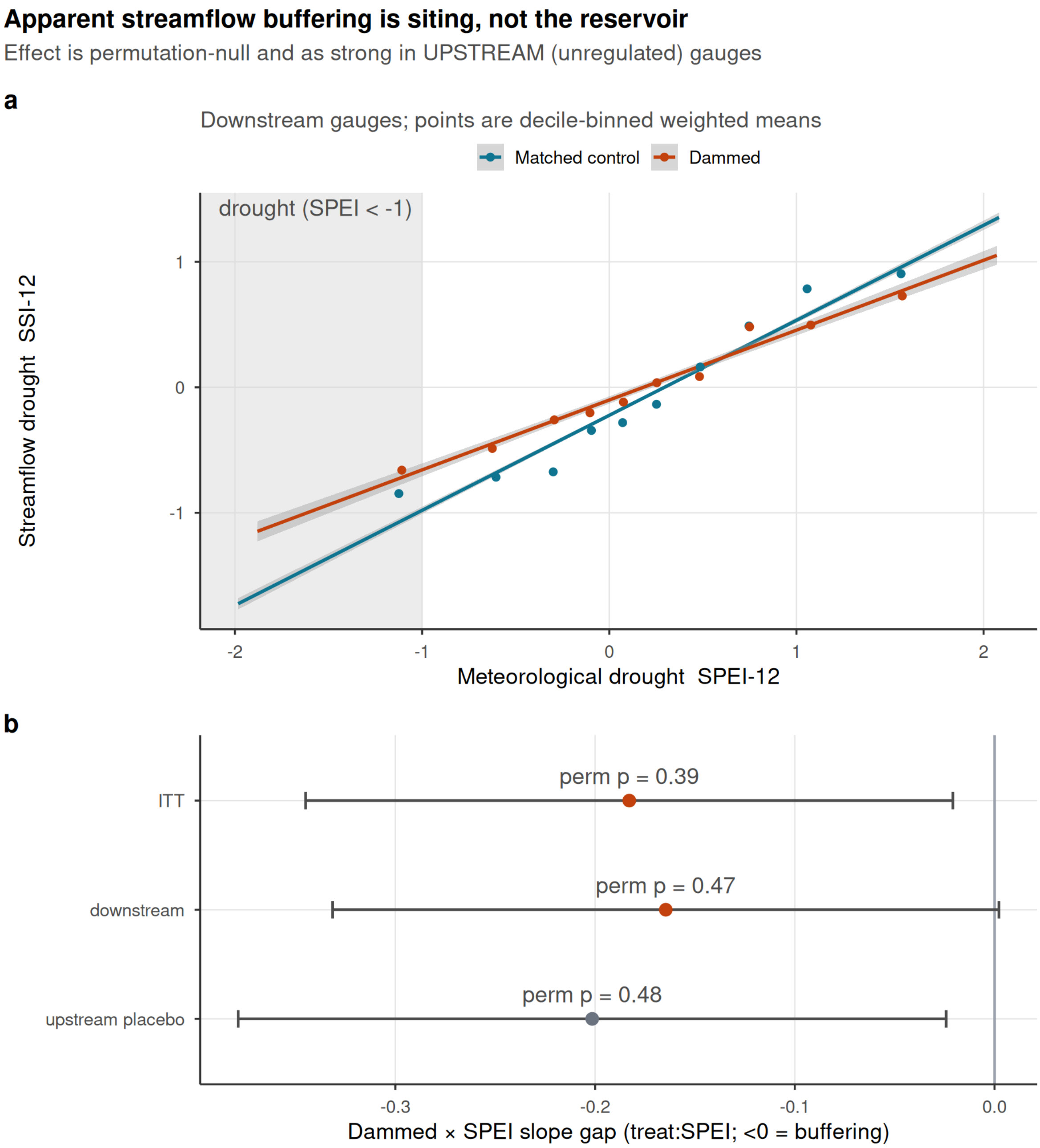

Figure 2: Streamflow buffering is reservoir siting, not regulation. (a) SPEI-12 to SSI-12 transmission for dammed versus matched-control basins (downstream gauges); lines are weighted least-squares fits and shaded bands are 95% confidence intervals; dammed basins show a flatter slope, the apparent buffering. (b) Differential slope (treat:SPEI; <0 = buffering) for intent-to-treat, downstream-only, and the upstream-only placebo; horizontal bars are 95% confidence intervals (estimate +/- 1.96 cluster-robust SE), annotated with randomization-inference p-values. All are permutation-null, and the upstream placebo (above-dam, unregulated) is as strong as downstream, so the attenuation is siting/aridity, not the dam.

buffering. Restricting evapotranspiration to vegetated (irrigated orchard) cells was an exploratory, post-hoc analysis prompted by that pre-trend failure, not a pre-specified outcome, so we read it as diagnostic of a mechanism rather than as an independent test: masking the barren pixels removes both the pre-trend failure (p from 3×10^{-5} to 0.40) and the positive effect (+0.048 to −0.020), consistent with the whole-basin signal being barren-land aridity contamination rather than a reservoir effect (Supplementary Fig. S7). Because the orchard stratum was defined after inspecting the whole-basin result, we rest no claim on it and note the general risk that post-hoc outcome redefinition can favour a desired result (Discussion).

This null is informative, not merely underpowered. With about 21 dammed clusters the design cannot claim exact equivalence to zero, but it rules out streamflow buffering larger than 46 to 58% of baseline transmission, and a positive control confirms the point: when we inject a known buffering effect into the treated gauges, the estimator recovers it one-to-one and randomization inference detects it once it exceeds that threshold (Methods; Supplementary Fig. S2 and Tables S1, S3). The design would therefore have detected a large operational effect had one existed.

Table 1: Reservoir-effect estimates across estimators. Cross-sectional doubly-robust matched ATTs (including the water-rights induced-demand test) and forcing-interacted DiD slope gaps (treat:SPEI). The DiD rows and water-rights accrual are judged by randomization-inference p-values (p), the design's primary inference at ~21 clusters; cluster-robust CIs are shown for completeness and over-reject here. A positive estimate would indicate induced-demand vulnerability; a negative one, operational buffering. *Whole-basin ET fails its event-study pre-trend test (p approximately 1e-10): its cluster-robust CI excludes zero but the panel is confounded by barren-land aridity, so it is labelled confounded and excluded from the equivalence bounds (Supplementary Table S1).

Group	Outcome	Estimand	Estimate	95% CI	p	Verdict
Cross-sectional matched ATT	Cropland-area expansion	ha km^-2 added	-0.687	[-2.44, 1.06]	-	null
Cross-sectional matched ATT	Reservoir ET buffering	d log ET / d SPEI	-0.014	[-0.0646, 0.0369]	-	null
Cross-sectional matched ATT	Water-rights accrual	rights / 100 km^2 added	20.036	[4.02, 36]	0.52	null
Forcing-interacted DiD (slope-gap)	Cropland area	differential deficit->impact slope	0.001	[-0.00344, 0.00493]	0.91	null
Forcing-interacted DiD (slope-gap)	Whole-basin ET	differential deficit->impact slope	0.048	[0.0255, 0.0703]	0.11	confounded*

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Group	Outcome	Estimand	Estimate	95% CI	p	Verdict
Forcing-interacted DiD (slope-gap)	Orchard ET	differential deficit->impact slope	-0.020	[-0.0448, 0.00565]	0.24	null

3.4. *Reservoirs mark, not make, agricultural vulnerability*

Dammed basins do carry the vulnerability reservoirs are blamed for concentrating: they hold about 4.6 times more cropland than matched controls (11.1% versus 2.4% of basin area; Figure 4, panel a). But this gap is a *siting* signature, present and stable across the entire 2000–2024 record, with dammed and control trajectories evolving in parallel through the megadrought. A dynamic event study confirms it: the differential trend is flat before 2010 ($p = 0.65$) and does not diverge afterwards ($p = 0.81$; Figure 4, panel b). Reservoirs mark where cropland already was; they neither expanded it nor changed how it responds to drought.

The same holds for the most direct measure of demand. In the DGA registry of consumptive water rights, dammed basins do appear to accrue rights faster than controls through the megadrought, on both the outlier-robust count (+20 rights per 100 km^2; Figure 3, Table 1) and the winsorized volume (+2.5 l/s per km^2; Supplementary Fig. S6), each with an analytic interval excluding zero. But this is the naive-analysis trap the design exists to catch: dammed basins are markedly more arid, arid basins carry denser water-rights activity, and this residual imbalance drives the gap. The count does not survive randomization inference ($p_{\text{perm}} = 0.52$), lying inside the null that random assignment within climate and aridity strata already produces (Supplementary Fig. S6), and it remains null under a fully non-parametric match restricted to the aridity-overlap subset ($p_{\text{perm}} = 0.42$; Supplementary Table S7). The winsorized volume is more delicate. It is stable across winsorization thresholds (95th to 99.9th percentile; Supplementary Table S8), yet on the overlap subset its stratified-permutation p falls to 0.005, apparently signalling an effect. That signal is an artefact of spatial dependence. Water-rights residuals are strongly spatially autocorrelated (Moran's $I = 0.39$, $p = 0.001$; Supplementary Table S9), which the unit-level stratified permutation does not preserve; under a spatially-restricted permutation that swaps treatment at the cuenca-block level the volume effect is no

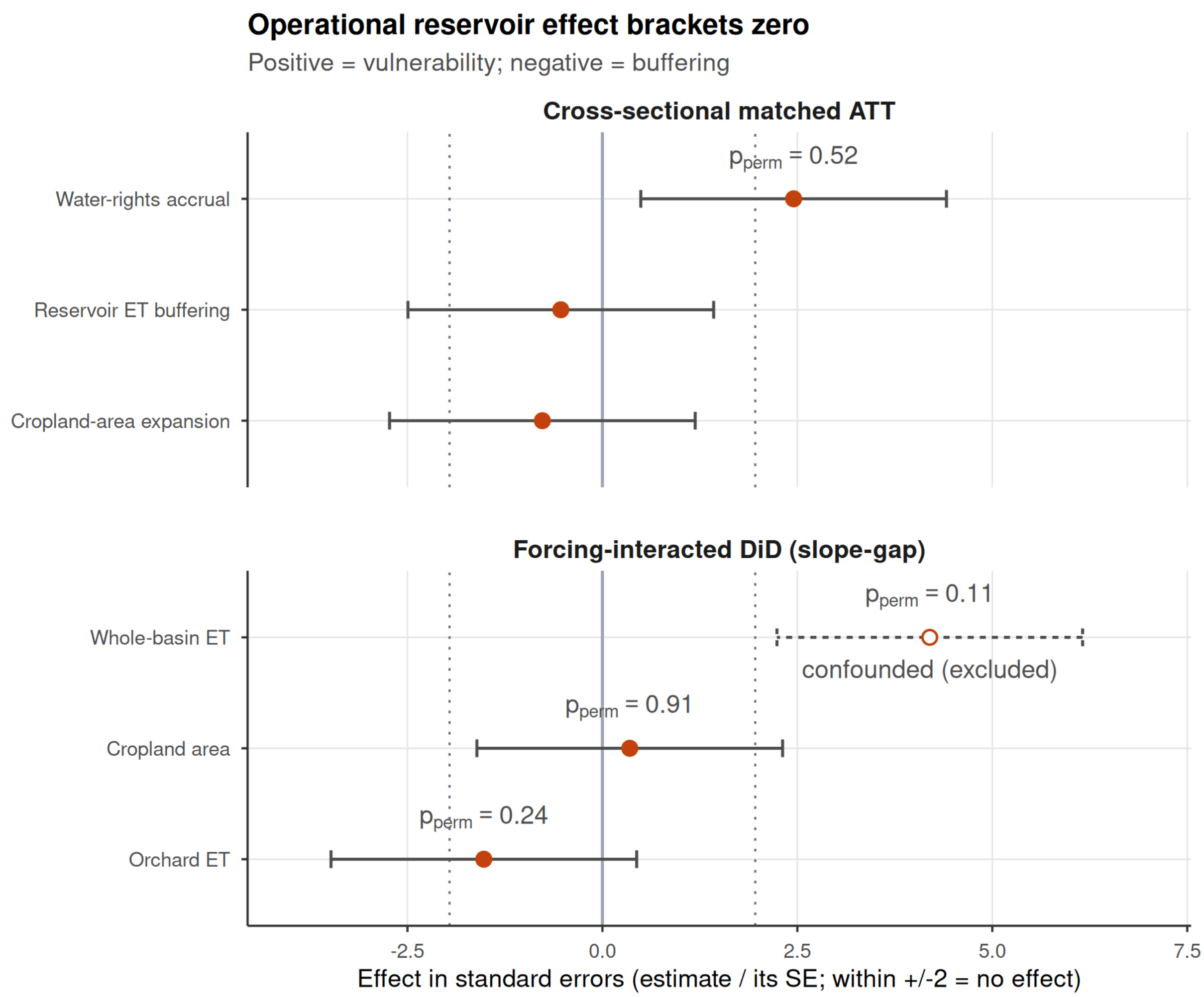

Figure 3: Dammed-versus-control effect across the estimators, each expressed uniformly in its own standard errors (estimate divided by its SE) so the axis metric is identical for every row; an estimate inside +/-2 is indistinguishable from no effect. Rows are grouped by inference regime: cross-sectional matched ATTs (top) and forcing-interacted DiD slope gaps (bottom). Bars span +/-1.96 SE (the 95% reference interval) and dotted lines mark +/-1.96; a positive effect would indicate induced-demand vulnerability, a negative one operational buffering. rows judged by randomization inference are annotated with their p (the design's valid inference at ~21 clusters, since the +/-1.96 SE guides over-reject there). Under that valid inference every estimate brackets zero. Two rows sit outside +/-1.96 by their naive standard error yet are null by the design: water-rights accrual (p_perm = 0.52; the apparent induced demand lies inside the permutation null expected from the dammed basins' unbalanced aridity, Supplementary Fig. S6) and whole-basin ET (open symbol, dashed interval), which additionally fails its event-study pre-trend test and is confounded by barren-land aridity, so it is labelled confounded and excluded (Table 1).

longer significant ($p_{\text{perm}} = 0.077$) and its cuenca-clustered interval spans zero ($[-0.3, 4.1]$). Even on water rights, then, no induced demand is robustly attributable to the reservoir once siting and spatial structure are accounted for, though a conventional matched analysis, or a spatially-naive one, would have concluded otherwise.

3.5. Storage falls; a change in refill amplitude cannot be resolved

If reservoirs neither buffer nor amplify drought, the change that matters under the megadrought is in how much water there is to store (Figure 5). Over 2005–2024 the entire storage band drifts downward in step: annual peak storage falls by 1.3 percentage points of capacity per year ($p = 0.013$) and the trough almost identically (1.2 pp yr^{-1}, $p < 10^{-4}$), while the seasonal amplitude shows no detectable trend (-0.06 pp yr^{-1}, $p = 0.91$; Supplementary Table S5). That amplitude interval is wide, so we cannot exclude a modest change in refill efficiency and do not claim the buffer is provably intact; what the band shows unambiguously is that it drops without clear narrowing. This descriptive trend is commensurate with the climate-driven supply reduction measured independently in the matched controls: unregulated streamflow in undammed control basins fell about 2.9% per year over 2005–2020 (dammed basins 3.4%), a relative rate comparable to the ~1.4% per year decline in reservoir peak storage. The comparison is correlative, not a naturalized-inflow or release-record analysis, so we cannot exclude a contribution from evolving management or unrecorded extractions, but the storage decline is at least consistent with, and no larger than, the falling supply seen in unregulated flow. Our direct demand outcomes point the same way: cropland area in dammed basins did not expand (Figure 4, Table 1), and neither did consumptive water-rights allocation (Supplementary Fig. S6) (Li et al., 2023; Wang et al., 2024).

4. Discussion

Across a matched national design and four independent estimators, and on the direct streamflow-availability outcome, the operational effect of reservoirs on drought brackets zero once siting is matched out. Dammed basins do hold about 4.6 times more cropland than matched controls, but that gap is fixed across the record, with parallel pre-trends and no divergence through the megadrought. The evidence supports a single reading: reservoirs **mark** where water-dependent vulnerability already concentrated, in arid, already-cultivated basins, rather than **make** it. The vulnerability signature is a property of siting, set before the drought and stable through it, not a dynamic consequence of regulation.

This is a finding, not a failure of power, because a credible *cause* of the apparent effect is identified: if dams

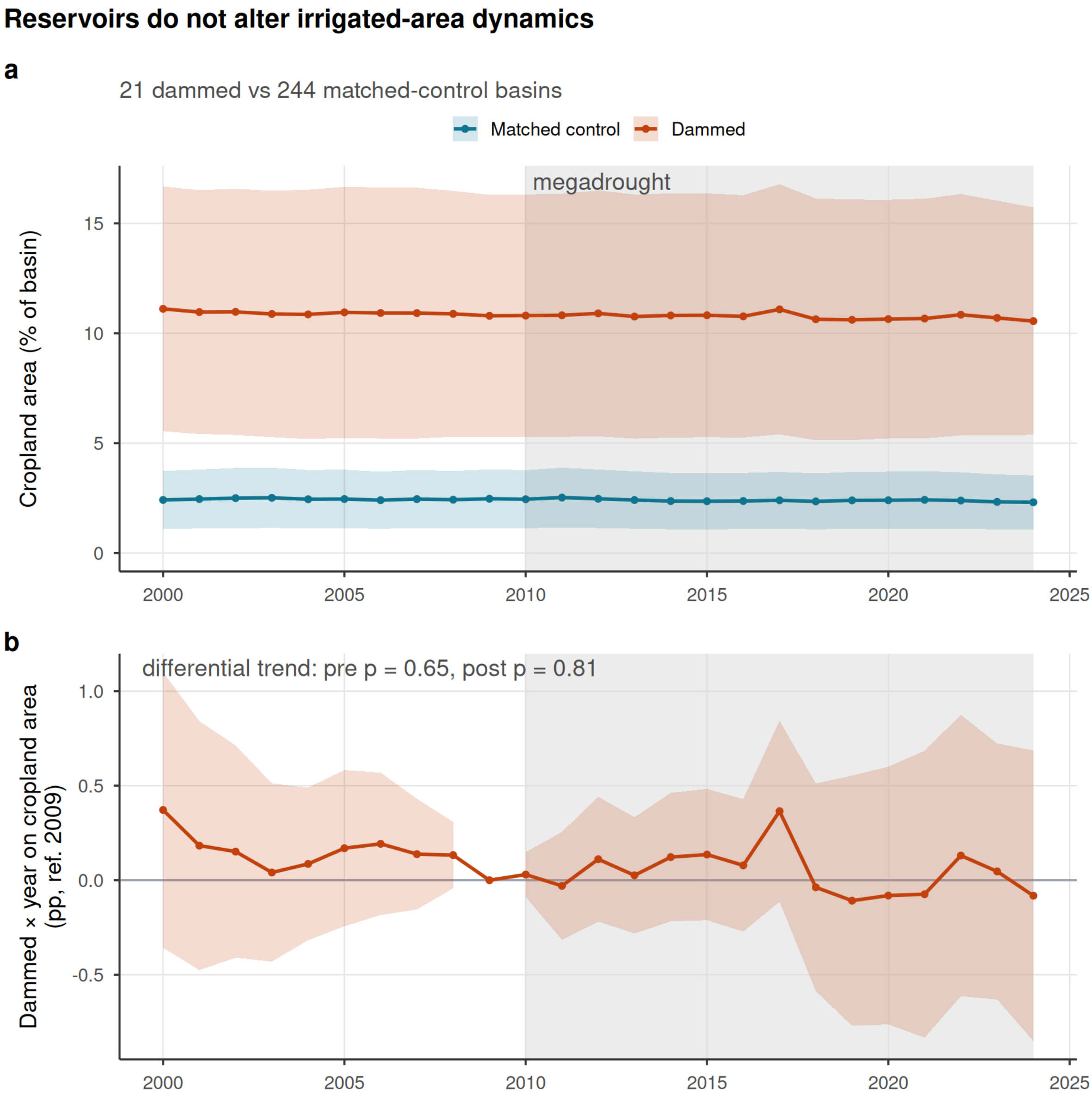

Figure 4: Observed cropland-area dynamics in dammed versus matched-control basins (21 dammed, 244 controls). (a) Entropy-balancing-weighted mean cropland area (% of basin, MapBiomas class 18) with 95% bands (weighted mean +/- 1.96 SE across basins); the y-axis is anchored at zero. Dammed basins hold ~4.6x more cropland (11.1% vs 2.4% of basin area) but the trajectories evolve in parallel. (b) Dynamic event study (dammed x year, reference 2009): the shaded band is the 95% confidence interval (coefficient +/- 1.96 cluster-robust SE). The differential pre-trend is flat (pre-2010 trend-slope $p = 0.65$) and does not diverge afterwards (post $p = 0.81$); a linear trend-slope test is used because the joint event-study Wald over-rejects at ~21 treated clusters. The 2010-2024 megadrought is shaded in both panels.

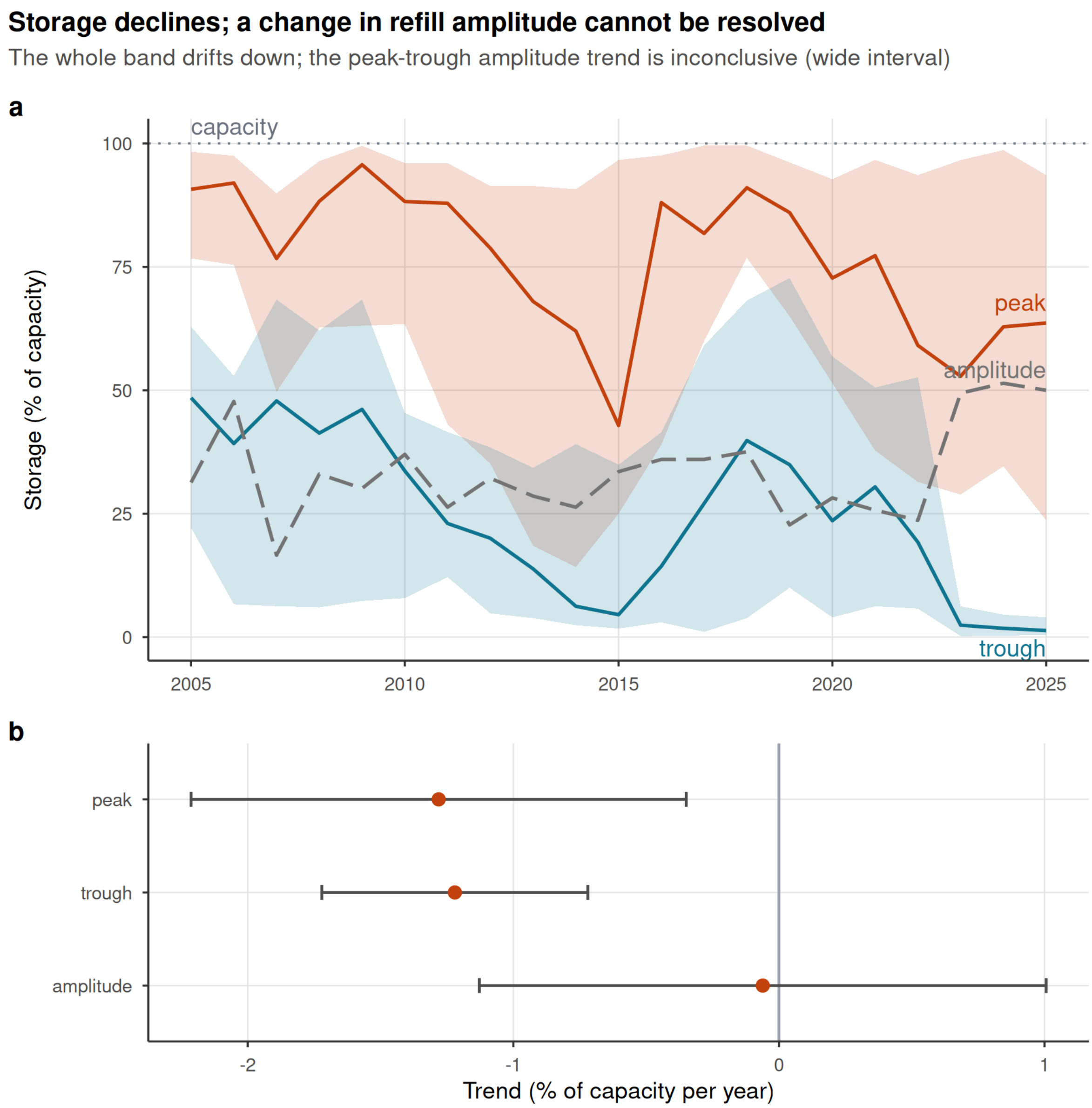

Figure 5: Reservoir storage declines; a change in seasonal refill amplitude cannot be resolved. (a) Cross-reservoir annual storage band: realized median peak and trough percent-of-capacity (solid lines) with interquartile ribbons across reservoirs, and the peak-trough amplitude (grey dashed). Medians are used because a few reservoirs report storage above nominal capacity (dotted 100% line); the whole band drifts down over 2005-2024 while the amplitude does not narrow. Central lines are the realized medians, not linear fits; the net trend is quantified in (b). (b) Per-year trend slope of peak, trough, and amplitude (reservoir fixed effects); error bars are 95% confidence intervals (estimate +/- 1.96 cluster-robust SE): peak and trough both decline while the amplitude trend is not distinguishable from zero (a wide interval, so statistically unchanged rather than proven constant), consistent with a supply-side level decline rather than refill/buffer degradation. Descriptive: pooled within-reservoir trend, no control group.

did what they are built to do our design was powerful enough to detect it, as the positive controls demonstrate, so the null refutes a core policy assumption rather than reflecting blindness. The within-basin placebo is the cleanest falsification: a genuine regulation effect must appear below the dam and be absent above it, and it is not; what remains, once the placebo and within-region conditioning are applied, is the aridity gradient itself. The corollary is general and transferable: a naive dammed-versus-control or calendar-time contrast reads the large cross-sectional gaps as a reservoir effect, whereas conditioning on realized forcing and benchmarking against an unregulated within-basin reach makes the signal disappear, so studies comparing dammed against undammed basins or pre- against post-construction levels inherit the full siting confound (Lee et al., 2026; López-Moreno et al., 2009; Schilstra et al., 2024; Wu et al., 2017); the upstream placebo applies to any regulated river with gauges above and below the structure.

If reservoirs neither buffer nor amplify multi-year drought operationally, the change that matters under a multi-decadal dry spell is in the water available to store. The storage record is suggestive: the whole storage band drifts downward, peak and trough declining in step at roughly 1.3 and 1.2 percentage points of capacity per year, while the seasonal amplitude shows no detectable trend (though its wide interval leaves a modest change in refill efficiency unresolved, so we do not claim the buffer is provably intact). A band that drops without clear narrowing points to a falling supply ceiling rather than a degraded buffer. This storage trend is descriptive (Condon et al., 2020); it is commensurate with the climate-driven supply reduction measured independently in the matched control basins, where unregulated streamflow fell about 2.9% per year over 2005–2020, a relative rate comparable to the storage decline. This comparison is correlative, however, not a naturalized-inflow or operational-release analysis, so we cannot rule out a contribution from evolving management practices or unrecorded upstream extractions; we therefore present the falling-supply reading as the most consistent interpretation rather than an identified cause. Our direct demand outcomes weigh the same way against a demand explanation: neither cropland area nor the allocation of consumptive water rights in dammed basins expanded differentially once siting is matched out. Read against Chile's documented transition from episodic drought toward structural aridification (Alvarez-Garreton et al., 2021; Garreaud et al., 2017; Vicente-Serrano et al., 2022; Zambrano et al., 2025) and the global pattern of reservoirs failing to refill and diminishing storage returns from new dams (Li et al., 2023; Wang et al., 2024), the more likely reading is that the binding constraint is shifting toward water supply rather than storage capacity or operation.

These findings speak to three international debates. They are the drought-domain expression of the socio-hydrological *reservoir effect* and *supply-demand cycle*, contested for want of quantification (Di Baldassarre et al., 2018; Kellner, 2021): we provide a credibly identified test and find the protective function over-attributed to storage. One scope condition matters: because treatment is time-invariant during a continuous drought, we test the *spatial* imprint of these feedbacks, whether they have already produced a vulnerability gradient across basins, not the longitudinal, coevolutionary loop (post-dam abundance to demand growth to heightened exposure) the theory describes, which would need staggered commissioning and longer records to observe. The findings also parallel the *efficiency paradox* in irrigation, where nominal water savings expand consumptive use (Grafton et al., 2018), and align with the falling marginal returns of new dams as reservoirs increasingly fail to refill (Li et al., 2023; Wang et al., 2024). The common lesson, that supply-side infrastructure does not resolve a scarcity driven by demand or falling supply, is strongest on the supply side; the demand-side reading is conditioned on a governance regime like Chile's that cannot curtail rights during scarcity, and where allocation is state-managed and adaptive the paradox need not follow.

The policy implication is direct and uncomfortable. Because Chile's 1981 Water Code makes water-use rights perpetual private property the state cannot readily curtail during drought, the reflexive response has been supply-side, a national programme of new dams and inter-basin transfers (Barría et al., 2021; Garreaud et al., 2025; Macpherson et al., 2023). The 2022 reform of the Water Code (Law 21.435) begins to soften this rigidity: it prioritizes water for human consumption, caps the duration of newly granted rights, and strengthens the state's authority to limit and reallocate abstraction during severe scarcity (Macpherson et al., 2023). Because the reform grandfathers the large stock of perpetual pre-2022 rights and postdates most of our study window, it does not affect our estimates, but it moves Chilean governance toward precisely the demand-side, adaptive allocation our results argue for; whether it curbs the demand expansion that a market regime invites is an important target for continued monitoring. For Chile and other regions facing structural aridification, our evidence indicates that building more storage is a supply-blind response to a supply problem: it cannot manufacture the water that aridification is removing, and the buffering benefit dams are built to provide is, on this evidence, a property of where they sit rather than of what they do. Adaptation must shift toward demand management, reallocation, and managed aquifer recharge rather than new impoundments (Grafton et al., 2018; Rivera-Vidal et al., 2025; Scanlon et al., 2023).

Several limitations bound these conclusions. Treatment is effectively time-invariant (most dams predate the record), so we identify differential drought *response* rather than the effect of switching a reservoir on; a staggered-commissioning design would be cleaner where construction dates and longer records permit. The result is a bounded null, not a proven equivalence: with ~21 clusters the design excludes only *large* buffering (>46–58% of baseline transmission for streamflow), so the causal weight rests on the placebo (Section 3.2), not on equivalence (Section 3.3). The observed-area outcome captures total cropland, not irrigation status, so a compositional shift toward water-intensive perennials could raise consumptive use without enlarging area, although the direct water-rights allocation, which would register such intensification as new claims, also shows no differential accrual once siting is matched out (Supplementary Fig. S6). Groundwater substitution is a related unobserved confounder: pumping that compensates for falling reservoir storage could sustain cropland and ET and so *mask* rather than reflect operational buffering; the groundwater water-rights series shows no differential accrual either, but realized pumping is unmeasured. Because dammed basins are markedly more arid, they may also draw their aquifers down differently from the matched controls through the megadrought, and the natural way to control for such differential depletion is satellite gravimetry (GRACE/GRACE-FO terrestrial-water-storage anomalies) or the DGA well-level network. GRACE's coarse footprint (~300 km) is larger than most of our subcuencas, so a basin-resolved gravimetric control is not yet feasible at this grain; assimilating downscaled storage products or well hydrographs to bound differential pumping between treated and control basins is a priority for future work, and until then we carry unmeasured groundwater as a subsidy that could support the demand-side nulls. The within-basin placebo carries its own residual confounders beyond the elevation gap we bound explicitly (Section 3.2): upstream reaches are steep, snow-dominated Andean headwaters whereas downstream reaches sit in flatter alluvial valleys, so differences in snowmelt storage and release, groundwater-surface-water exchange, and channel geomorphology could shape the upstream-versus-downstream transmission slope in ways elevation alone does not summarize. These differences would have to act *differentially* between dammed basins and controls to manufacture the null, and they cut both ways (snowmelt storage, for instance, is itself a natural buffer that would tend to flatten the upstream slope and so *mimic* rather than mask a reservoir effect), but we cannot fully exclude them and flag the upstream/downstream contrast as comparability-limited rather than a perfect natural experiment. Our evapotranspiration outcomes inherit a further product-level caveat: MOD16 infers ET from meteorology and vegetation indices and does not represent the anthropogenic water applications that decouple soil moisture from local precipitation, so it is known to underestimate ET over heavily irrigated land in arid settings, exactly the orchard cells we analyse. Such underestimation would most plausibly *compress* the deficit-to-impact slope where irrigation sustains canopies through drought, biasing the orchard-ET transmission gap toward zero and so toward our null; a positive induced-demand signal masked by MOD16 cannot be ruled out and would need a water-balance or thermal ET product to resolve. The orchard stratum itself was defined post-hoc, after the whole-basin ET pre-trend failed, so we treat its null as a mechanistic diagnostic of barren-land contamination rather than a confirmatory test and place no independent weight on it, mindful that data-driven outcome redefinition can otherwise inflate the chance of a convenient result. Inter-basin transfers, part of Chile's supply-side programme, are a further spillover pathway that our within-cuenca contamination rule does not capture; any treated-control pair linked by a transfer would violate spatial independence, though such linkage would tend to bias the matched contrast *toward* the null we report. On the timescale, the 12-month aggregation targets annual-to-interannual transmission, not the sub-seasonal (1–3 month) buffering that irrigation reservoirs are primarily built to provide and may well still deliver, so our conclusions concern long-term, multi-year vulnerability under structural aridification, not the short-term operational resilience that we neither test nor dispute. Finally, the operational finding, that buffering reflects siting not regulation, should hold wherever dam siting is non-random, but the demand-feedback and policy inferences are conditioned on Chile's unusually market-based, weakly adaptive governance; under state-managed allocation, operators able to curtail demand or re-time releases could in principle produce genuine buffering. A remaining possibility we cannot fully adjudicate is that damming leaves the magnitude of impact unchanged while recomposing its *dependence structure*, substituting a storage-to-outcome coupling for the meteorology-to-outcome one. A concrete design would exploit **rule-curve deviations** as a quasi-experiment: operating rules set target storage as a function of season and antecedent state, so the residual between actual releases and the rule-prescribed release is a source of variation in delivered water that is orthogonal to contemporaneous inflow. Regressing downstream outcomes on that release residual (instrumented by rule-curve targets, with basin and year fixed effects) would identify a genuine storage-to-outcome coupling separately from the forcing-to-outcome coupling we estimate here; the data requirement is reservoir-release and rule-curve records, which exist operationally but are not yet public, and any decoupling so identified must still survive the upstream placebo. Assembling these release records, alongside longer streamflow series and water-rights transaction histories, is the priority for future work.

Returning to the question that motivates the study, whether reservoirs create resilience or merely postpone vulnerability under prolonged drought, our evidence indicates that at the multi-year timescale of structural aridification they do neither operationally: over annual-to-interannual accumulations storage neither buffers nor amplifies the drought signal beyond what siting already implies, while the water available to store steadily declines. This verdict concerns the multi-year horizon our design resolves and does not contest the intra-annual, seasonal buffering that irrigation reservoirs are primarily built to provide, which our 12-month accumulation is not designed to detect. Resilience to a falling supply ceiling will come not from more impoundments but from confronting demand and allocation, the harder and unavoidable adaptation.

5. Data and code availability

The full analysis pipeline, comprising the `targets` workflow, the reusable R functions (`src/R/`), the study-parameter configuration (`config/`), and the scripts that generate every figure and table, is openly available at https://github.com/frzambra/dam_drought_effectiveness and archived on Zenodo (DOI: 10.5281/zenodo.XXXXXXX). The pinned R environment (`renv.lock`) reproduces the package versions used. Reservoir storage records, the derived result tables, and matched-set metadata are included in the repository; the third-party input datasets are publicly available from their providers as cited: CHIRPS/CHIRTS (SPEI-12 forcing and baseline aridity), MODIS MOD16 (evapotranspiration), MapBiomas Chile (land cover), the CR2 network (streamflow), SRTM (elevation), and the DGA reservoir, watershed, and water-rights registries. Every result in the manuscript corresponds to a named pipeline target, so the analyses can be regenerated with a single `targets::tar_make()`.

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